

# TeamSpace Farm: Nurturing innovative agricultural technology with data and cloud

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TeamSpace Farm startup founded with the goal of using data to change the agricultural environment. Describing the beginning on a small scale, it created a data-driven environment through big data and machine learning.

## About TeamSpace Farm

TeamSpace Farm is a company that creates a new agricultural environment where anyone can grow agricultural produce through data and technology in our everyday environment.

Centering on intelligent IoT environmental sensors that analyze and process data in real time, they are setting the foundations to achieve Artificial Intelligence-driven automatic control agriculture, utilizing big data.

Industries: Agriculture, Forestry & Fishing

Google Cloud, an AI (https://cloud.google.com/vertex-ai/), developed a machine learning-based sensor device, all ready to use with the business TeamSpace Farm. The rewards of data by predicting and crop growth in its current tomato and medical hemp fields.

Location: Korea

AI (https://cloud.google.com/vertex-ai/),

Google Cloud (https://cloud.google.com/)

(https://cloud.google.com/looker-studio/)

BigQuery (https://cloud.google.com/bigquery/)

Vertex AI (https://cloud.google.com/vertex-ai/)

Looker Studio

(https://cloud.google.com/looker-studio)

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## Google Cloud results

- Integration of machine learning model learning, service, and deployment environment with Vertex AI
- TPU-based Coral development

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Build professional machine learning-based data workflows in a small development environment and deploy sensor and compute

board  
simplifies  
data  
processing  
procedures

- Providing a  
data-driven  
smart farm  
platform

integrated  
hardware at one-  
tenth the cost

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Agriculture is one of the oldest industries in human history. But looking back, there hasn't been much change in the fundamental ways in which the environment and the corresponding experience are controlled.

Why is the possibility of smart farm business that data and IT technology will change the agricultural environment attracting attention? Because it is not different from the direction of digital transformation that higher value can be obtained by extensively and delicately handling the vast amount of data pouring out in the farming environment.

"TeamSpace  
Farm started  
with the goal of  
building a data-

based  
agricultural  
environment.  
Although we are  
a small startup,  
Google Cloud  
presented an  
integrated  
environment that  
covers the entire  
workflow from  
sensors and  
hardware to  
cloud and  
machine  
learning, so we  
were able to  
focus on  
developing  
services without  
worrying about  
infrastructure  
and  
development  
platforms."

The story of TeamSpace Farm

(<https://spacefarm.co.kr/en>), which was founded with the aim of changing the agricultural environment based on crop data and eventually, farming in space if conditions can be properly controlled, is fascinating in its ambition.

## **Smart farming and migrating to data agriculture**

With the world's population still growing rapidly and our quality of life dependent on reliable and quality food resources, the demand and interest in smart farms is huge. Despite this, the reality of agricultural fields is still limited to first-generation technologies, such as sensor checks and remote environmental control.

However, agriculture and IT technology are looking forward to the next step in real-time monitoring of the growth of crops based on data and artificial intelligence, and predicting appropriate response methods and corresponding yields. By focusing on this, TeamSpace Farm realized that handling data well can improve the quality of farming.

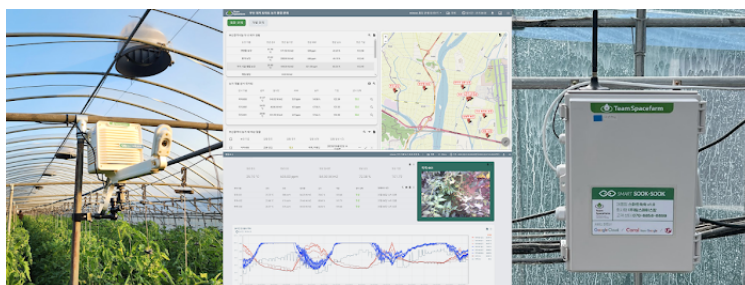
"It is not easy to change fundamental methods in farming. This is because the damage caused by a moment's mistake is enormous. Most of the farming process is made by nature, including sunlight and water. It is difficult for humans to adjust the optimal conditions through changes in the environment.

However, I believe that farming can be done anywhere, including indoors or in containers, when precise environmental monitoring and an artificial

intelligence-based growth prediction model based on it are completed."

Yu-ri Seo, CEO of TeamSpace Farm, said that he decided to start the business after realizing that he could control the environment desired by crops through data. "I chose Google Cloud as the basis for quickly turning that possibility into reality. After all, I believe that the key to smart farms is to effectively collect and process a lot of data, and quickly distribute responses to the results."

TeamSpace Farm provides farmhouses with a smart farm solution that monitors the environment with an integrated IoT smart sensor in an agricultural environment where conditions are controlled, such as a greenhouse, and identifies how crops are actually growing based on the data.



*Greenhouse installation in Daejeo Area*

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"If it weren't for Google's services, it wouldn't have been easy to achieve results in the field right from the start. It is impossible for a newly founded startup to directly build all the IT workflows. However, Google presented the entire workflow from the hardware to the cloud data pipeline, and we were

able to focus on developing targeted ideas based on the features we needed."

Soungno Kim, CTO of TeamSpace Farms, put connectivity and consistency at the center of the initial conception of the service. In the process, it was determined that Google Cloud connects all processes from the sensor to the cloud most smoothly. It was judged that dividing the infrastructure, such as multi-cloud or hybrid cloud, makes management difficult and interferes with the flow of data. And in fact, Google was open to various possibilities based on a wide ecosystem. A typical example is sensor hardware installed in agricultural fields.

The 'Smart Growing Plant' sensor developed by TeamSpace Farm is a device that is installed in indoor farms, including plastic greenhouses. This single sensor checks temperature, humidity, atmospheric pressure, light intensity, and carbon dioxide in real time, and also manages the growth stage of crops through a camera. This sensor is the core and starting point of smart farms, but most smart farm sensors are sensor nodes and integrated controllers. In many cases, each additional element is built separately, so sensors and equipment often increase. The cost is tens of thousands of won, and the facilities are complicated because there is so much equipment. However, TeamSpace Farm was able to supply a small computer-based integrated sensor for around 2 million KRW. This is because the development board of 'Coral,' a subsidiary of Google, was used.

In particular, this integrated development board contains a small computer optimized for machine learning and data processing, such as GPU and TPU, in addition to basic sensors. The data obtained through the sensor can be directly processed and judged by the sensor itself, not a separate computer.

## **Configuring a consistent data workflow from sensor to cloud**

Collected data is primarily processed in the sensor. The TPU inside the device analyzes anomalies that need to be quickly identified in the field, such as pests, through pre-trained machine learning models, and the results are sent to Google Cloud through the network. Because of this pre-processing process through GPU and TPU, on-site response speed is faster and more accurate, while the burden of transmission, processing, and storage of the entire data is drastically reduced. The data pre-processed by the sensor flows to BigQuery in Google Cloud.

"BigQuery is a unique service with no substitutes. Data coming from IoT sensors is primarily used in Cassandra storage for immediate dashboarding rather than analytics. As the Cassandra database grew in size, it was difficult to manage with limited manpower. Currently, Cassandra is used only for real-time monitoring, and the full-fledged data is sent to BigQuery for storage and analysis."

"BigQuery not only serves as a warehouse that integrates and manages all data required by TeamSpace Farm, but also serves as a window for users to access data in agricultural fields." Seongno

Kim, CTO, emphasized that he was very satisfied with BigQuery in all aspects, including capacity, processing speed, and cost.

The core asset of TeamSpace Farm is a machine learning model that can determine the growth of each crop based on the data accumulated in BigQuery. All of these machine learning models are created and managed through Google Cloud infrastructure, including Vertex AI and Google Compute Engine. TeamSpace Farm focuses on three machine learning models: prediction, suggestion, and Vision AI. Learning is continuously taking place, and reinforcement learning is performed with field data even during actual farming. As machine learning models are improved or created, they are sent back to IoT sensors remotely.

In this way, Vertex AI goes beyond simply creating each machine learning model, instead its learning the model, applying the service, and even distributing the machine learning model created and distributing it to a smart sensor. All processes related to machine learning are integrated in one place. Seongno Kim, CTO, explains that thanks to the integrated artificial intelligence environment, various machine learning attempts can be continued.

"The multi-spectral camera attached to the smart sensor takes pictures of the crops to check if they are growing properly. This camera can read infrared information, and it can read the points that the main components of the physiological action of crops respond to specific infrared bands. This image information is interpreted through GPUs and TPUs

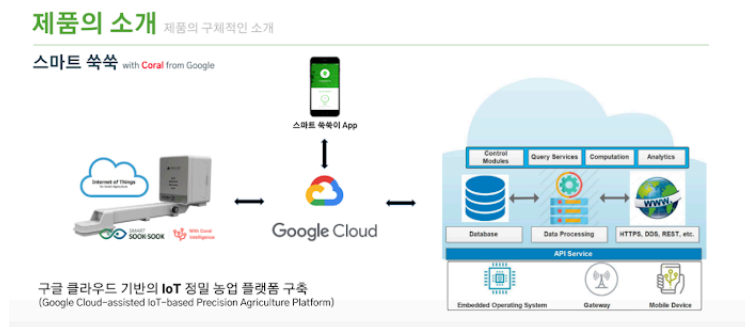
within the smart sensor, and machine learning models based on Tensor Flow."

TeamSpace Farm applied a model that can judge whether farming is going well by analyzing not only indoor environment data but also photos of crops through Vision AI. As environmental information and growth information are collected at the same time, it is possible to predict plant growth based on this data. A machine learning model that determines the size and shape of a leaf just by taking a picture can immediately recognize growth and pests and diseases. We are also developing a machine learning model that can figure out how big the fruit will grow and predict the quality of the fruit, such as yield and sugar content, as data accumulates, and we plan to launch the service soon.

Artificial intelligence technology is also used to accurately measure temperature and humidity. Since the temperature and humidity sensors are inside the case, heat may be generated from the processor, and since it is difficult to directly measure the outside air, errors may occur with the actual temperature. This, though, is also corrected through machine learning based on real-time conditions inside the device.

Currently, based on this technology, results are being produced in the Daejeo Tomato Complex in Busan and the Medical Hemp Special Zone in Andong. Machine learning models for each crop can be created in a matter of weeks, making it possible to apply the product 'Smart Sook-Sook,' which is an

agricultural IoT environmental sensor service to virtually any crop.



*Google Cloud-assisted IoT-based Precision Agriculture Platform*

## Google Cloud enables start-ups with limited manpower and resources to dream

Seongno Kim, CTO, explains that the greatest strength of Google Cloud is that it can turn all these possibilities into reality. In fact, being able to create a service that can have an impact in the real world with little resources and as a new business that started with challenges and ideas is actually the most important part. The advantage of a startup is that it becomes easier to tell the next story only when you can see the value of the company with your own eyes.

"The biggest advantage I feel when using Google Cloud is 'manpower.' There are currently 3 engineers at TeamSpace Farm. These three people are doing everything from hardware to infrastructure, machine learning to platform development. At first, there were

times when we thought that the journey to start the service would not be easy if we did not have enough staff."

From the beginning of service development, CTO Kim Seong-no decided to completely change the way we work with the existing ones. "We decided to boldly abandon the traditional method of building all infrastructure and making service elements ourselves. Instead, if there is a reliable service from outside, the strategy is to actively utilize it. To complete this, the public cloud was the key, and we decided that in order for all services to operate organically within it, we should handle all of them in one place without dualizing the platform. Google Cloud was very well equipped with all conditions, from data collection and storage, processing and distribution, which were the core of smart farms, to hardware modules."

TeamSpace Farm was able to build an integrated IoT sensor and data platform that could be put on the market in just 1 year and 6 months.

CTO Kim Seong-no emphasized that the strength of Google Cloud is an environment where founding members with early ideas can test their ideas and turn them into products, even while the system is operated and developed, and new services are developed.

"Google Cloud is developer-friendly. All services are optimized to produce optimal functions. It is easy to manage and set up, and the Google Cloud CLI-based commands allow developers unfamiliar with Google

Cloud to easily utilize all of the features of Google Cloud. Overall, it is an environment where the cloud environment and the developer's development environment are tightly integrated."

Of course, there were difficulties with using public cloud services. Initially, TeamSpace Farm's IoT sensor was developed through Google's IoT Core service. However, shortly thereafter, IoT Core was shut down and TeamSpace Farm was embarrassed by the sudden end.

"Google and its partner Megazone actively helped right after the IoT Core service termination announcement. Google Korea helped me choose a better alternative by providing various data on alternative services. Engineers from Google and Megazone not only provided technical support, but also helped us through the migration process, so we were able to move to the new service without difficulty."

It started by finding a technical alternative, but in fact, the concern of companies using public cloud, PaaS, and SaaS services is also the 'sustainability of the provided service.' However, since better services are still being developed, as long as it is not a simple relocation and finding an alternative, you can create a better experience by directly transferring to another service as needed, rather than for external reasons, such as terminating the service.

**Dreaming of changing the way we farm with data**

TeamSpace Farm's next goal is to change the way farming is done. CEO Yu-ri Seo is convinced that the long-established agricultural environment can find a better way around data. Smart Farm is an inexpensive and relatively simple technology, it is easy to access, and it can make numerous decisions while farming throughout the year by handling data.

"I want to deliver a new data-driven agriculture to farmers. Data related to crops can be easily checked, and through this, changes in the farming method in which decisions are made can be made. Google Cloud makes it easy to show data to service users as well."

Access to data through Looker Studio, which shows BigQuery information as BI, creates a starting point for farmers to experience new ways of data farming through real-time dashboards, helping them handle BigQuery as needed, or easily obtain better insights through Google Spreadsheets.

TeamSpace Farm's idea is that once data-based farming technology is established, farming can be done in space. Beyond looking at space, exploration is actually happening. The possibility that mankind can live in space presents a solution to the food problem, and eventually bringing farming to space is a true pioneering story.

CEO Seo Yu-ri believes that dealing with food comes with responsibility. CEO Yu-ri Seo expresses his aspirations to make changes in the agricultural market through a clear philosophy and vision that benefits many people widely, rather than simply

providing technology. Google Cloud has become part of this journey with Google sharing TeamSpace Farm's vision of helping the world with technology, not just services.

"TeamSpace Farm is a company that handles food. We want to be a company that develops technology to make farming more effective, and furthermore, play a role in creating an equal and free world where food is available to all. I think this philosophy fits well with the idea that Google helps people benefit with technology. We will create a better world together with Google, based on easy-to-use artificial intelligence technology."